



INTERNSHIP & JOB PLACEMENT HUB

TECHNICAL SPECIFICATIONS

Project Manager:

Muhammad Motara (38008149)

Group members:

Student number	Name	Surname
38700220	Everhardus Louw	Hugo
38136198	Divan	le Grange
45645892	Precious	Masetla
41595270	A	Montjane
40923894	Hardus	Myburgh
38177005	Junior	Ndlovu

38178877	Erlin	Nghondzweni
38008106	Cynthia	Ngwari
43511139	Ronm	Nwariwe
38647745	Bohlale	Nyedimane
21052239	Christo	Prinsloo
44396244	Tumelo	Same
24962724	Sd	Steyn
41993128	Morena	Themane

1. Preface	4
1.1 PURPOSE	4
1.2 INTENDED AUDIENCE AND PERTINENT SECTIONS	4
1.3 PROJECT SCOPE	5
1.4 REFERENCES	6
2. DESCRIPTION	7
2.1 PRODUCT PERSPECTIVE	7
2.2 FEATURES	7
2.3 USER OVERVIEW	7
2.4 OPERATING ENVIRONMENT	7
2.5 CONSTRAINTS: IMPLEMENTATION / DESIGN	7
2.6 DOCUMENTATION	7
2.7 ASSUMPTIONS / DEPENDENCIES	7
3. SYSTEM FEATURES	7
3.1 Adding Records	7
3.2 Search and Filtering	7
3.3 Folder Management	7
3.4 Profile Management	7
3.5 Showing Current Market Value of Records	7
4. REQUIREMENTS OF EXTERNAL INTERFACE	7
4.1 USER INTERFACES	7
4.2 HARDWARE INTERFACES	7
4.3 SOFTWARE INTERFACES	7
4.4 COMMUNICATION INTERFACES	7
5. ADDITIONAL NONFUNCTIONAL REQUIREMENTS	7
5.1 PERFORMANCE	8
5.2 SAFETY and SECURITY	8
5.3 SOFTWARE QUALITY	9

1. Preface

The Job and Internship System has been developed to address the challenges faced by students and graduates in accessing career opportunities, as well as the difficulties employers experience in finding suitable candidates efficiently. Many graduates struggle to discover and apply for relevant jobs or internships, while employers often face lengthy and inefficient recruitment processes.

This platform provides a centralized, structured environment where students and graduates can create profiles, explore and apply for opportunities, and track their applications. Employers can post job and internship listings, review candidate applications, and gain insights into hiring trends. By integrating these features into one system, it enhances employability, supports workforce development, and strengthens the connection between education and employment.

1.1 PURPOSE

The purpose of this Technical Specifications Document is to provide a detailed outline of the technical foundation, architecture, and design decisions for the Job and Internship System. While the functional specification describes what the system should do, this document focuses on how it will be implemented.

It serves as a guide for all project stakeholders, including developers, database engineers, testers, project managers, and end-users, to understand the technologies involved, integration methods, and responsibilities required for development and deployment. This document ensures that all stakeholders have a shared understanding of the system's technical expectations.

By clearly defining the technical requirements, security measures, operating environment, and system architecture, the document establishes a reference point for integration, scalability, and security considerations. This ensures that the platform is built to handle current requirements while remaining adaptable for future growth, enabling informed decisions about maintenance, upgrades, and overall system performance throughout the project lifecycle.

1.2 INTENDED AUDIENCE AND PERTINENT SECTIONS

Developer's Crucial Areas

Front-end and back-end developers, as well as database engineers, will rely on this document to understand the technical structure of the Job and Internship System. Key areas of focus include:

- Technical Specifications: Detailed descriptions of the technology stack, databases, server architecture, and programming frameworks used to implement the system.

- Software Interfaces: Guidance on integrating internal modules and external APIs, ensuring seamless communication between the system components.
- Operating Environment: Requirements for hardware, operating systems, and browser compatibility to guarantee smooth operation.
- Security Requirements: Implementation of authentication, authorization, encrypted communication, and data privacy measures to comply with relevant regulations such as POPIA.

Project Manager's Crucial Areas

The project manager will use this section to ensure that technical work aligns with project goals, timelines, and quality standards:

- Project Scope: Overview of system objectives, functionalities, and technical boundaries.
- Constraints: Information on any limitations related to technology, budget, or schedule that could impact delivery.
- Assumptions and Dependencies: Clarification of external dependencies, required tools, and assumptions for risk management and planning.

User's Crucial Areas

End-users, including students, graduates, and employers, will not need to implement the system but benefit from understanding its capabilities:

- User Interfaces: How the system should behave, look, and interact from the perspective of students, graduates, and employers.
- Features: Description of available functionality, including profile management, job/internship postings, search and application workflows, notifications, and reporting tools.

1.3 PROJECT SCOPE

The Job and Internship System is designed to provide a centralized platform that connects students, graduates, and employers efficiently. The scope of this project includes the design, development, and deployment of a web-based system with the following capabilities:

Candidate-Facing Features:

- Account creation and secure sign-in for students and graduates

- Profile management including uploading CVs, skills, and academic records
- Browsing and searching of job and internship opportunities
- Application submission and tracking
- Notifications and alerts for new opportunities and application status

Employer-Facing Features:

- Account creation and secure sign-in for employers
- Job and internship posting with detailed descriptions and requirements
- Viewing, reviewing, and managing candidate applications
- Communication with candidates regarding application status
- Access to reports and analytics on applications and candidate engagement

System-Wide Capabilities:

- Secure login and role-based access control
- Data storage and management for users, postings, and applications
- Integration with external services and APIs if needed
- Analytics and reporting tools for admins to monitor system performance and usage
- Compliance with data protection regulations such as POPIA

The system is intended to streamline the hiring process, improve the visibility of opportunities for students and graduates, and provide actionable insights for employers and administrators. While the system will be web-based, basic mobile access may be available, though mobile optimization is not a core requirement at this stage.

1.4 REFERENCES

Internship&JobPlacementHubBC&FS.docx

Internship&JobPlacementHubTechnicalSpecifications.docx

Internship&JobPlacementHubFunctionalSpecifications.docx

2. DESCRIPTION

2.1 PRODUCT PERSPECTIVE

In South Africa, a lot of graduates leave university with qualifications but then struggle to get into the job market. One of the big issues is that their CVs are often not standardised, sometimes missing key details, or even hard for employers to trust. On the other side, companies spend time going through piles of CVs that don't always give them what they need.

At the moment, most graduates use Word/PDF CVs, emails, or platforms like LinkedIn and job boards. The problem is that none of these are really standardised or verified. The Graduate Placement Portal improves on this by bringing everything into one system: a professional profile, verified documents, employer access, and an application tracker.

The idea behind the Graduate Placement Portal is to fix this gap by giving graduates one place where they can create proper profiles, upload and verify their documents, and apply for jobs directly. Employers also benefit because they can quickly find reliable information instead of digging through unreliable CVs. The portal will be web-based and cloud-hosted so that graduates, employers, and institutions can access it from anywhere. It's designed to scale up as more users join. In other words, it takes a process that is currently scattered and unreliable, and replaces it with a single, trusted platform.

Put simply, the portal is meant to make life easier: graduates can focus on presenting themselves well, and employers can make faster and better hiring decisions.

2.2 FEATURES

- **Profile & CV Management:** Build and update a professional profile online.
- **Document Verification:** Upload certificates/transcripts and get them verified.
- **Employer Access & Search:** Employers can search and filter candidates.
- **Application Tracking:** Students can track applications; employers can track responses.
- **Notifications:** Automatic alerts for job postings or updates.
- **Data Security:** Encrypted storage, POPIA-compliant, role-based access.

2.3 USER OVERVIEW

Primary Users – Students and Graduates

- Mostly final-year students or recent graduates.
- Different levels of computer skills (some advanced, some basic).
- Eager to start their careers and need a way to show their qualifications clearly.

What they need:

- A simple platform to create/update a profile.
- A safe way to store and verify documents.
- The option to apply for jobs directly without extra steps.

Secondary Users – Employers / Recruiters

- HR staff, hiring managers, or recruitment agents.
- Don't have time to sort through unverified or inconsistent CVs.
- Need a reliable way to filter and find the right candidates.

What they need:

- Quick candidate search and filtering tools.
- Easy access to verified qualifications.
- A smoother way to track applications.

Future Users – Institutions and Career Services

- Universities and colleges wanting to see how their graduates perform in the job market.
- Career centres that support students with placements.

- Alumni groups tracking long-term career outcomes.

2.4 OPERATING ENVIRONMENT

- **Hardware compatibilities:**

The web-app we will be building does not have a specific hardware platform we are aiming it to function on. Most reliable web browsers should be able to access the website and interact with the necessary features (such as Google Chrome), whether that be on a Personal Desktop Computer or a smartphone.

- **Operating system compatibilities:**

Again, since the application will be web-based, the worry about specific OS or software requirements are less important. We will be aiming to build our web-app in such a way that any reliable operating system using a reliable web browser will be able to access the website.

- **Database requirements:**

A cloud-hosted service like AWS/Azure/GCP/Firebase Hosting will be used to host the database, with Firebase Firestore or PostgreSQL used to manipulate the database data.

- **Front- and backend environments:**

Node.js or Firebase functions will be used for the backend and React.js or Flutter for the frontend.

- **GitHub and other collaboration platforms:**

When it comes to building, testing and deploying the system, we will be using platforms such as GitHub to build and host the codebase for our project. This provides version control, collaboration and other useful features. WhatsApp and Discord will also be used to communicate casually in order to schedule meetings and other correspondence.

2.5 CONSTRAINTS: IMPLEMENTATION / DESIGN

- **Technical difficulties:**

Our technical challenges will be important to overcome. Learning new languages and API's, as well as new technologies such as cloud hosting and online databases, will pose big challenges for us in this project and overcoming them will be the important way in which we successfully complete this project.

- **Limited Time:**

We only have a limited time to finish this project. Designing, building, testing and implementing all the necessary features will take time to do and completing all these steps for all the components of the project before the deadline will be important in order to deliver a polished product.

- **Budgetary restrictions:**

We have no budget for this project. Everything will need to be done on our own dime if we need to pay for hosting databases or other services. This may be an issue if we do not find a way to work around or solve it.

- **Distance format:**

Since we are mostly working from home/online and it is unrealistic to work together in person, our proper coordination and planning, as well as cohesive teamwork will be important in ensuring we stay up to date with our schedule and that all the features are properly designed, built and implemented.

2.6 DOCUMENTATION

- **Timeline Documentation:**

Documentation will be maintained throughout the development process in order to ensure that all the functional parts are satisfied and that the project development timeline remains clear. Deadlines and goals will be set and aimed to be completed at certain intervals.

- **Code documentation:**

We will use inline code commenting to explain the working and functionality of the written code during the development process. Naming conventions will also be established to ensure readability and proper collaboration.

- **Database schema documentation:**

We will provide the schema and specifications for the database in order to ensure clarity regarding the working of the database. This includes the used tables, relationships, etc.

- **User Manual:**

A simple feature guide will be added to explain the different features and how to use them. This is aimed at the general user in order to provide an idea of how to interact with the system.

- **Standards and delivery:**

In order to produce a polished product that will have market appeal and adoptability, we will be adhering to user experience and interface clarity standards. This includes industry best practices in developing the system.

2.7 ASSUMPTIONS / DEPENDENCIES

- **Adoption rates:**

We assume that businesses will be interested in placing advertisements for job opportunities on our site and that users will be willing to browse our site and apply for these jobs, since we will be providing a streamlined and convenient way of doing so. The current market lacks a platform/service such as ours and thus we assume good adoption rates.

- **Internet and browser access:**

We assume that our clients will have internet access and access to reliable web browsers in order to interact with our site.

Our clients will be dependent on their internet access to use our site.

- **API and software integration:**

We assume that the existing API's and software we will be relying on to build, host and interact with our site will be up to the task of providing all the expected features. Our project will be dependent on these technologies to be continued to be supported into the foreseeable future.

- **Time constraints:**

We assume that we will have enough time to complete our project.

Our successful completion of the project relies on our ability to handle any unforeseen problems that may arise.

3. SYSTEM FEATURES

Create Job/Internship:

DESCRIPTION	Allows employers to create a new job or internship listing.
EVENT SEQUENCE	Action: Employer clicks on the "Create Listing" button. Inputs: job title, description, required skills, location, employment type, and application deadline. Processing: The system displays a form, validates the input, and saves the post in the database. Outputs: Confirmation message, listing is visible to prospective applicants.

Edit Job/Internship:

DESCRIPTION	Allows employers to edit an existing job or internship listing.
EVENT SEQUENCE	Action: Employer clicks on “Edit Listing” button. Inputs: job title, description, required skills, location, employment type, and/or application deadline. Processing: The system displays a form for changing details, then validates and saves the post in the database. Output: The system displays a form to input listing details, validates the input, and saves the listing.

Delete Job/Internship:

DESCRIPTION	Allows employers to delete an existing job or internship listing.
EVENT SEQUENCE	Action: Employer clicks on “Delete Listing” button. Inputs: Selection of the job or internship listing to delete. Processing: The system prompts for confirmation, removes the listing from the database, and updates the employer’s dashboard. Output: The system displays a form to input listing details, validates the input, and saves the listing.

Job/Internship Matching

DESCRIPTION	Provides searching and filtering capabilities for students to apply for relevant job or internship opportunities based on their skills, field of study, and location.
EVENT SEQUENCE	Action: Student clicks on the search bar. Inputs: Search criteria such as job title, location, skills, or field of study. Processing: The search is performed based on criteria. Outputs: Results are displayed.

Submit Application

DESCRIPTION	Students can submit applications for Jobs/Internships.
EVENT SEQUENCE	Action: Students click on “Apply” button for a job/internship post. Inputs: Required documents (CV, cover letter, portfolio if applicable). Processing: The system prompts the student to select, or upload required documents. Outputs: Confirmation message, application is saved and visible to the employer.

Track Application Status

DESCRIPTION	Students can monitor applications for Jobs/Internships.
EVENT SEQUENCE	Action: Student accesses the dashboard. Inputs: Student account identifier and selected application. Processing: The system retrieves the current application status. Outputs: The application is displayed and visible to the student (received, accepted, rejected).

User Registration

DESCRIPTION	Allows new users (students or employers) to create an account on the application.
EVENT SEQUENCE	Action: User clicks on “Register” on the homepage. Inputs: User details such as full name, email, password, account type (student or employer). Processing: The system displays a registration form, validates input, and creates a new account in the database. Outputs: Confirmation message, user is able to log in

User Login

DESCRIPTION	Allows existing users to log in to the application.
EVENT SEQUENCE	Action: User clicks on “Login” on the homepage. Inputs: Email/username and password. Processing: The system validates credentials against stored accounts. Outputs: User is granted access to his/her dashboard if username/password combination is correct, or an error message if not.

User Profile Management

DESCRIPTION	Allows users to view and update their profile information after logging in.
EVENT SEQUENCE	Action: User clicks on “Edit Profile” on the dashboard. Inputs: Updated profile information (e.g., name, contact details, skills). Processing: The system displays the current profile in an editable form, validates changes, and saves the updated information. Outputs: Confirmation message, profile is updated and changes reflect in the user account.

Employer Verification

DESCRIPTION	Allows administrators to verify the accounts of employers to ensure that only legitimate employers can post job or internship listings.
EVENT SEQUENCE	Action: Employer submits verification documents on their profile. Inputs: Verification documents such as business registration, company ID, or other required credentials. Processing: The system stores the submitted documents and notifies administrators for review. Administrators validate the documents and approve or reject the employer account. Outputs: Confirmation message upon submission. Employer account is approved or rejected. If approved, the employer is assigned permissions to post, edit and delete job/internship listings.

4. REQUIREMENTS OF EXTERNAL INTERFACE

4.1 USER INTERFACES

Style Guides and Standards

- **Platform Guidelines:** The portal will follow Google's **Material Design principles** to ensure a modern, intuitive, and visually consistent design.
- **Standards:** The interface will adopt **responsive web design**, ensuring usability across desktops, tablets, and mobile devices, regardless of orientation or resolution.

Constraints

- **Device Compatibility:** The UI must adapt to different device types (desktop browsers, Android/iOS mobile browsers).
- **Accessibility:** Text-to-speech compatibility, proper colour contrast, and keyboard navigation must be supported.

Functionality

- **Navigation:** Clear navigation bars, role-based menus (Student, Employer, Administrator), and breadcrumb trails for ease of movement between modules.
- **Interactivity:** Features like dropdown filters, toggle buttons, file upload drag-and-drop, and instant feedback messages (e.g., "CV uploaded successfully").
- **Search and Filters:** Advanced search (by skill, location, company, or field of study) with sorting and filtering options to streamline browsing job opportunities.
- **Notifications:** Real-time updates for application status, interview requests, and employer messages.

Sample Screens

- **Login Screen** (role-based access: Student, Employer, Admin)
- **Registration Screen** (account creation with secure authentication)
- **Home/Dashboard** (customized for role: students see opportunities, employers see applications, admins see stats)
- **Browse/Search Opportunities** (filters by skill, field, or location)
- **Application Screen** (upload CV, apply, track status)
- **Employer Post Job Screen** (job description, requirements, deadlines)
- **Applicant Management Screen** (employers filter and shortlist candidates)
- **Admin Panel** (approve postings, monitor analytics, manage users)

4.2 HARDWARE INTERFACES

Features of the Product / Hardware Interfaces

1. Compatibility with Devices:

- **Desktops/Laptops:** Accessible via Chrome, Firefox, Safari, and Edge.
- **Mobile Devices (Android/iOS):** Fully functional mobile browser access; future native mobile app support is possible.

2. Requirements for Operating Systems:

- **Cross-Platform Support:** As a web-based portal, the system runs independently of OS (Windows, macOS, Linux, Android, iOS).
- **Browser Support:** Minimum Chrome , Firefox , Safari , Edge .

3. Utilization of Resources and Performance:

- Optimized for **low latency** and **minimal CPU/memory usage**, even on budget mobile devices.
- Cloud hosting ensures scalable performance during peak loads (e.g., graduate recruitment season).

Nature of Interactions

1. User Interface Interactions:

- **Touch & Click Inputs:** Users interact via taps, swipes (mobile), or clicks (desktop).
- **Form-Based Input:** Upload CVs, complete application forms, post job listings.
- **Dashboard Widgets:** Interactive charts and counters for quick statistics.

2. Multimedia Interactions:

- **Document Handling:** Upload and view resumes in PDF/DOC formats.
- **Visual Displays:** Company logos, job banners, and student profile photos.

4.3 SOFTWARE INTERFACES

Components

1. Operating systems:

- Must have cross-platform support as it is web-based.

2. Databases:

- Database containing active and inactive job listings as well as user data.

Constraints

Must meet all requirements while keeping to the budget set for this project. Must be able to correctly display on all platforms to ensure user access to the site. A stable internet connection is important, to ensure up to date job, employee and employer information.

Communication and Data Flow Nature

1. Inbound data

- Data provided by users: User interaction provides data such as job, employee and employer information.

2. Outbound data

- User Data Synchronization: ensures up to date client data such as user information and job listings.

4.4 COMMUNICATION INTERFACES

Users will be need to login and register, as well as see a dynamic job listings board and messaging feature to allow them to communicate with employers of listings that the users are interested in.

5. ADDITIONAL NONFUNCTIONAL REQUIREMENTS

5.1 PERFORMANCE

Responsiveness:

The app should respond quickly when users navigate between pages, search for jobs, or upload documents. This will be achieved by proper database design, for example table indexing and query optimization. Targets of response times below 3000ms for page navigation and search results below 3s will be used to measure responsiveness.

Performance testing will be conducted on lower-specification devices including budget Android phones and entry-level laptops to ensure consistent user experience across all target devices.

Efficiency:

It should run smoothly on normal laptops, desktops, and mobile phones without needing very high system specifications. Overcomplicating the site must thus be avoided. Performance testing will be conducted on lower-specification devices including budget Android phones and entry-level laptops to ensure consistent user experience across all target devices.

Offline Functionality:

Users should be able to view already saved information (like their profile or previously seen job ads) even if the internet is down, with changes syncing later. Offline indicators will be used to inform users when connectivity issues arise.

System Performance:

The platform should allow multiple students and employers to use the system at the same time without slowing down noticeably. The web server must thus have adequate specifications to handle the projected web traffic for the website. Performance monitoring tools (APM) will track key metrics including response times, throughput, error rates, and resource utilization with automated alerting for performance degradation. We will test the application's performance with 10 users running at the same time. The MySQL database will also be set up using indexes to improve performance.

Error Handling:

If something goes wrong (e.g., failed login, file upload error), the system should show clear messages and guide the user on what to do next. Design principles must thus be implemented for the development to have similar approaches. Stress-testing must be performed to the error-handling capabilities. Error handling will be validated through chaos engineering and stress testing to ensure system resilience under failure conditions.

Scalability (Basic):

The system design should allow adding more features or supporting more users later without needing a full rebuild. This can be done by using web servers that are easily scalable and have little hardware change requirements.

5.2 SAFETY and SECURITY

Data Security:

User sensitive information including credentials, personal data, CVs, and applications must be protected using industry-standard encryption. Some penetration testing will be performed to ensure that the data is stored securely.

API Communication Security:

Any connection with other services (e.g., emails or notifications) should not expose sensitive information. Secure protocols like HTTPS must be used to protect data being sent and received.

User Authentication:

A login system must separate students, employers, and administrators so that each can only access their own features. To this end, authentication mechanisms will be implemented to prevent unauthorised access or data leaks. Role-based access control can be used for students, job posters etc.

Password Protection:

Passwords must not be visible to anyone, including system administrators. A simple hashing technique can be used to protect them. Relevant password methodologies must

also be applied, for example password policies will enforce minimum 8 characters with complexity requirements (uppercase, lowercase, numbers, special characters).

Session Management:

Users should be logged out after a certain period of inactivity to prevent unauthorized use. The system must thus log inactivity and log out a user after a certain period of inactivity.

Backup of Data:

Regular database backups should be done so that no important user data is lost if the system fails. This can be achieved by using an adequate cloud server provider that provides backup services. A recovery time objective will be set at 4 hours, in line with industry standards. Daily backups should be available for data restores.

5.3 SOFTWARE QUALITY

Intuitive Interface:

The application must provide an intuitive user experience following established UI/UX principles including progressive disclosure, consistent mental models, and clear visual hierarchy.

Task Efficiency:

Posting a job, applying for a position, or editing a profile should take only a few simple steps. This will also be achieved by applying user interface design principles like ensuring minimum data keying etc. Between 3 and 5 steps will be used as the baseline to determine the task efficiency.

Discoverability:

Important features such as “Apply,” “Upload CV,” or “Post Job” should be clearly visible and easy to find. This can be achieved by creating separate interfaces for job applicants and job posters and creating a central navigation pane.

Accessibility:

The system should work properly on both desktop and mobile devices. This can be done using Flexible Layout Techniques or Flexible Grid Systems. Cross-browser compatibility testing will ensure consistent experience across all major browsers and assistive technologies.

Consistency:

The same design style (colours, buttons, navigation) should be used across all pages, so users do not get confused. A design guidance must be provided to the development team to ensure a uniform approach to UI design.

Feedback to Users:

Whenever a task is completed (e.g., application submitted, job posted), the system should confirm it with a success message. Relevant notification and error warnings must be built into the code to give clear instructions when a user does something outside the scope of the user inputs.

Help and Guidance:

A small help section, tooltips, or instructions should be available to assist new users in understanding how to use the platform. This can be built into the user interface to be available in all screens. Help content will be role-specific and will be regularly updated based on user feedback.